

Cloudflare mTLS Step-by-Step Implementation

Overview

Production-style Cloudflare mTLS implementation for securing machine-to-machine traffic.

Architecture

Client → Cloudflare Edge → mTLS Validation → Worker/API

Step 1 — Create Client Certificate

Generate a client certificate and private key using internal PKI or OpenSSL.

Step 2 — Upload Certificate

Zero Trust → Access → Service Auth → Mutual TLS Certificates → Upload certificate.

Step 3 — Configure Hostname

Use a dedicated hostname such as mtls-test.volblut.com.

Step 4 — Create WAF Rule

```
(http.host eq "mtls-test.volblut.com"
and not (
  cf.tls_client_auth.cert_verified
  and cf.tls_client_auth.cert_issuer_ski eq "YOUR_SKI"
  and cf.tls_client_auth.cert_serial eq "YOUR_SERIAL"
))
```

Step 5 — Enable Certificate Forwarding

Enable client certificate forwarding for Workers/applications.

Production Recommendations

Use rotation, logging, per-vendor certificates, and replay protection.